Bereket Tilahun

Backend Security Engineer · Addis Ababa, Ethiopia
Fixed-price security reviews for AI-built Supabase, Next.js and Stripe apps on Upwork.
berakhah.github.io · github.com/Berakhah · upwork.com/freelancers/~01645da719eb4829ed

Experience

Senior Software Engineer May 2026 – Present
DZ Software Engineering PLC · Addis Ababa, Ethiopia
Promoted to Senior in May 2026; part-time since August 2026.

Software Engineer Jul 2023 – Apr 2026
DZ Software Engineering PLC · Addis Ababa, Ethiopia
Completed my BSc in Computer Science at HiLCoE alongside this role.

- Built and operated 15 containerized backend services on FastAPI and Django REST, shipping through CI/CD; layered Redis caching (84% hit rate) to drop p95 latency from 340ms to 195ms at 420 req/min peak.
- Architected an LLM evaluation pipeline and CI gate benchmarking correctness, reliability and runtime across Python, JavaScript, Java, C++ and Rust submissions — rolled out as an automated quality gate on 11 production services, cutting manual review by 40% with zero correctness regressions since launch.
- Designed an automated benchmark curation pipeline over PostgreSQL telemetry and Redis event streams, reducing MTTD by 30% and producing 1,200+ labeled records for fine-tuning.
- Hardened 47 API endpoints with schema-enforced validation, parameterized queries, rate limiting and Bandit/pip-audit pre-merge gates, closing 9 High/Critical findings from internal reviews and dependency scans with no recurrence.
- Replaced 11 long-lived secrets with short-lived, scope-limited OIDC tokens, made Trivy image scans a hard deploy gate, and stood up an append-only audit pipeline; incident reconstruction now takes under 20 minutes.
- Deployed evaluation services on AWS (EC2, S3, IAM role-scoped access, VPC-isolated mesh) behind load-balanced ingress, achieving environment parity across 11 staging/production pairs and reducing rollback time by 35%.

Full Stack Developer Mar 2022 – Jun 2023
DZ Software Engineering PLC · Addis Ababa, Ethiopia

- Triaged 10–12 Tier 1 incidents daily across Python, Java and C++ services; authored runbooks and automated recurring health checks, cutting recurring failures by 15% across enterprise environments.
- Diagnosed and resolved OS, network and distributed-service failures end-to-end, establishing observability baselines that were later adopted across 3 production environments.

Freelance Backend Engineer (part-time) Mar 2021 – Jun 2023
Independent, for individual organizations · Remote
Part-time backend work before and alongside DZ, until mid-2023.

- Built backends on FastAPI, Django REST and Spring Boot, with React/Node.js frontends.
- Engineered a multilingual code search backend with BM25 and TF-IDF retrieval; enforced authorization-before-query checks to neutralize timing-oracle attacks and applied AES-256 tenant-scoped encryption at rest.
- Folded correctness scoring, latency profiling and reliability checks into a unified CI/CD-gated pipeline, eliminating manual handoffs between frontend, API and backend stages.

Projects

- **ShopFast sample audit** (2026, demo, not client work): a demo store with 8 planted flaws (3 Critical, 2 High, 3 Medium), all 8 proven in a live run and 6 re-proven by one command, each with its fix.
- **Agent Perimeter** (2026, open source, Apache-2.0): security scanner for MCP servers and AI agents; 34 checks in 8 groups, 940 test functions (Nov 2026), and a census of 31,953 MCP Registry entries.
- **Ledger Sense self-audit** (2026, personal project, not client work): 34 findings (1 Critical, 15 High, 18 Medium) in my own invoice app. Reproduced and fixed the Critical cross-tenant export delete (LS-01); 11 of 16 Critical/High fixed, 5 open and listed.

Skills

Backend	Python 3.12 · FastAPI · Django REST · Spring Boot · Pydantic v2 · SQLAlchemy 2 + Alembic · PostgreSQL · async workers · event-driven pipelines · Typer CLI
Security	Zero Trust · OIDC / Keycloak · JWT & token lifecycle · RBAC / ABAC · OWASP Top 10 · threat modeling · SARIF 2.1.0 · envelope encryption · Argon2 · seccomp profiles · HMAC · Bandit / pip-audit · Trivy · gitleaks
Data	Redis · DuckDB · SQLite · Parquet · NumPy / SciPy / scikit-learn · BM25 / TF-IDF · pdfplumber · pypdfium2 · RapidOCR
Infrastructure	Docker & Compose · GitHub Actions · AWS (EC2, S3, IAM, VPC) · Linux · structlog · Prometheus · multi-stage builds · CI/CD gates
AI & Evaluation	LLM evaluation pipelines · cluster-BCa bootstrap intervals · Cohen's κ · Krippendorff's α · Benjamini–Hochberg · O'Brien–Fleming sequential testing · isotonic / Platt calibration · LLM-as-judge bias controls · hypothesis property testing

Education

B.Sc. Computer Science — HiLCoE School of Computer Science & Technology, Addis Ababa Jul 2026
Completed while working as a software engineer at DZ Software Engineering (2022–2026).